



Lecture 4

4.1 Complex Form of Fourier Series

*4.2 Uniform Convergence and Fourier Series

*4.3 Dirichlet Test and Convergence of Fourier Series

4.1 Complex Form of Fourier Series

フーリエ級数の複素数形式

4.1 Complex Form of Fourier Series

Complex Number

Review Complex Analysis

Imaginary Unit

$$i = \sqrt{-1} \quad \Rightarrow \quad i^2 = -1$$

Pure Imaginary Number

Define **pure imaginary number** as $z = bi$,

where b is a **real number** and i is the **imaginary unit**.

For example, $z = 6i$ or $z = -2i$ is a pure imaginary number.

A **complex number** is defined as $z = a + ib$, where a and b are **real numbers** and i is the **imaginary unit**.

4.1 Complex Form of Fourier Series

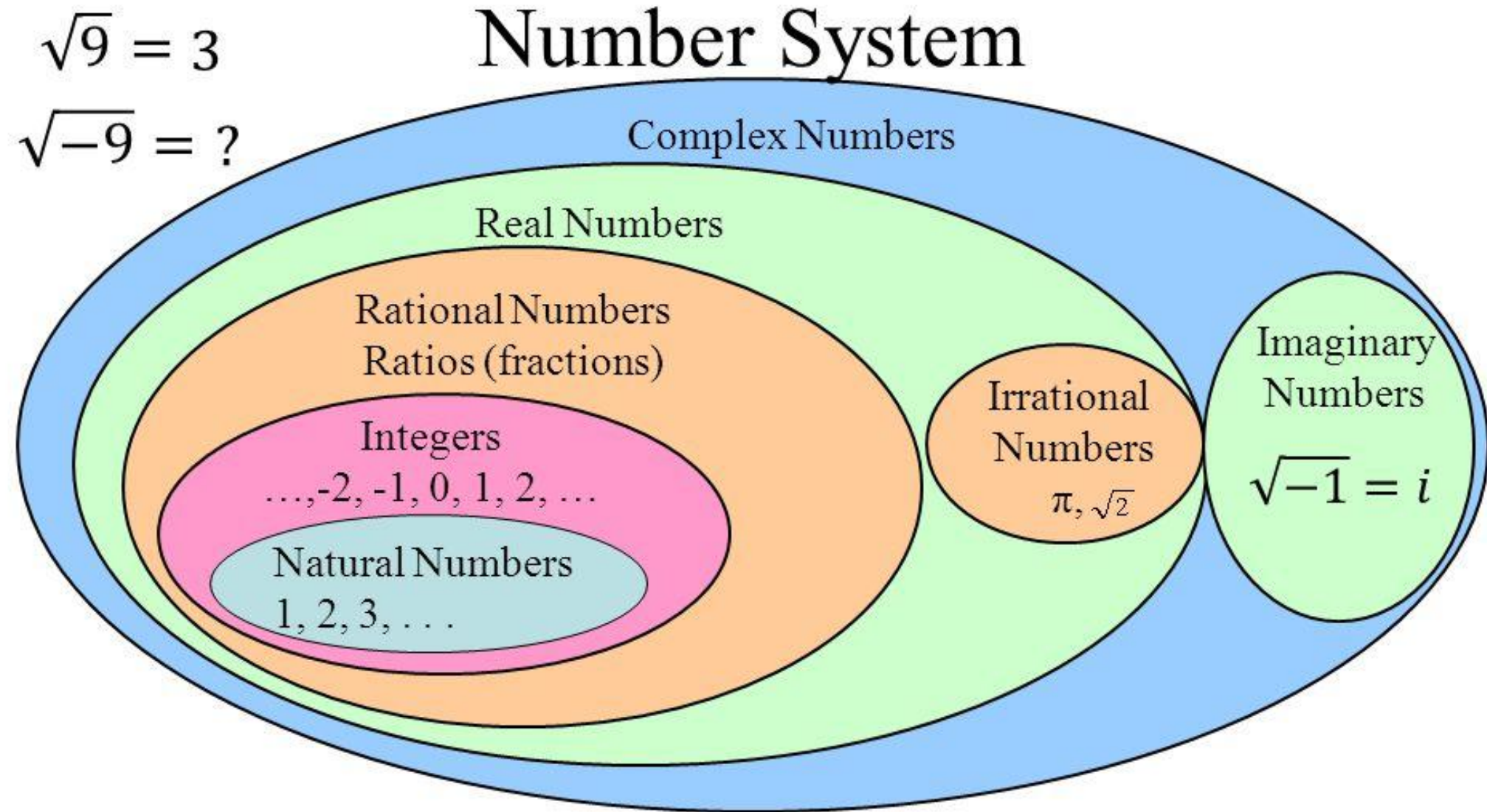
The set of Complex numbers

The set of Complex numbers is usually denoted by the symbol $\mathbf{C}$ or $\mathbb{C}$. i.e.

$$a + ib \in \mathbf{C}, \quad ib \in \mathbf{C}, \quad a + i0 \in \mathbf{C}$$

Notice: Because any real number a can be written as $z = a + i0 = a$, we see that the set $\mathbf{R}$ of real numbers is a subset of $\mathbf{C}$.

4.1 Complex Form of Fourier Series



Credit: <https://www.slideserve.com/ankti/complex-numbers>

4.1 Complex Form of Fourier Series

Real Part and Imaginary Part of Complex Number

$$\text{In } z = a + ib,$$

the real number a is called the **Real** part of z , i.e. $\text{Re}(z)$;

the real number b is called the **Imaginary** part of z , i.e. $\text{Im}(z)$.

For example:

$$\text{if } z = 4 - 9i, \text{ then } \text{Re}(z) = 4 \text{ and } \text{Im}(z) = -9.$$

Notice:

$$\text{Re}(z) \in \mathbf{R} \text{ and } \text{Im}(z) \in \mathbf{R}.$$

4.1 Complex Form of Fourier Series

Equality

If real numbers $a_1 = a_2$ and $b_1 = b_2$, then Complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ are equal, i.e. $z_1 = z_2$.

(Two complex numbers are equal if their corresponding real and imaginary parts are equal.)

If we use the symbols $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$, above definition states that

$$z_1 = z_2 \text{ if } \underbrace{\operatorname{Re}(z_1)}_{a_1} = \underbrace{\operatorname{Re}(z_2)}_{a_2} \text{ and } \underbrace{\operatorname{Im}(z_1)}_{b_1} = \underbrace{\operatorname{Im}(z_2)}_{b_2}.$$

4.1 Complex Form of Fourier Series

Review that in *Calculus II*, we have Taylor Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

4.1 Complex Form of Fourier Series

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$$

$$e^{ix} = 1 + \frac{ix}{1!} + \frac{(ix)^2}{2!} + \frac{(ix)^3}{3!} + \frac{(ix)^4}{4!} + \frac{(ix)^5}{5!} + \dots$$

$$= 1 + \frac{ix}{1!} - \frac{x^2}{2!} - \frac{ix^3}{3!} + \frac{x^4}{4!} + \frac{ix^5}{5!} - \dots$$

because $i^2 = -1$

$$= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) + \left(ix - \frac{ix^3}{3!} + \frac{ix^5}{5!} - \dots \right)$$

$$= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \right) + i \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right)$$

$$= \cos x + i \sin x$$

4.1 Complex Form of Fourier Series

$$\textcircled{1} \quad e^{ix} = \cos x + i \sin x \quad \text{Euler's formula}$$

$$\textcircled{2} \quad e^{-ix} = \cos(-x) + i \sin(-x) \\ = \cos x - i \sin x$$

$$\textcircled{1} + \textcircled{2} \quad e^{ix} + e^{-ix} = 2 \cos x \quad \Rightarrow \quad \cos x = \frac{e^{ix} + e^{-ix}}{2}$$

$$\textcircled{1} - \textcircled{2} \quad e^{ix} - e^{-ix} = i 2 \sin x \quad \Rightarrow \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i}$$



Leonhard Euler
(1707-1783)
Mathematician

4.1 Complex Form of Fourier Series

We know the trigonometric identity

$$\cos x \cdot \cos y = \frac{1}{2} [\cos (x + y) + \cos (x - y)]$$

Because $\cos x = \frac{e^{ix} + e^{-ix}}{2}$ $\sin x = \frac{e^{ix} - e^{-ix}}{2i}$ then we have

$$\begin{aligned}\cos x \cdot \cos y &= \frac{e^{ix} + e^{-ix}}{2} \cdot \frac{e^{iy} + e^{-iy}}{2} \\&= \frac{1}{2} \cdot \frac{e^{i(x+y)} + e^{i(x-y)} + e^{i(-x+y)} + e^{i(-x-y)}}{2} \\&= \frac{1}{2} \left(\underbrace{\frac{e^{i(x+y)} + e^{-i(x+y)}}{2}}_{\cos(x+y)} + \underbrace{\frac{e^{i(x-y)} + e^{-i(x-y)}}{2}}_{\cos(x-y)} \right).\end{aligned}$$

4.1 Complex Form of Fourier Series

For complex form of Fourier series , let us start with the two identities

$$\cos x = \frac{e^{ix} + e^{-ix}}{2} \quad \text{and} \quad \sin x = \frac{e^{ix} - e^{-ix}}{2i} \quad (4.1)$$

where $x \in \mathbf{R}$.

We will use these identities to find a complex form for the Fourier series expansion of a **2p-periodic** function

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x \right) \quad (4.2)$$

4.1 Complex Form of Fourier Series

Theorem 4.1 Complex Form of Fourier Analysis

Let f be a $2p$ -periodic piecewise smooth function. The complex form of the Fourier series of f is

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i \frac{n\pi}{p} x} \quad (4.3)$$

where the Fourier coefficients c_n are given by

$$c_n = \frac{1}{2p} \int_{-p}^p f(x) e^{-i \frac{n\pi}{p} x} dx \quad (n = 0, \pm 1, \pm 2, \dots) \quad (4.4)$$

For all x , the Fourier series converges to $f(x)$ if f is continuous at x , and to $\frac{f(x-) + f(x+)}{2}$ otherwise.

4.1 Complex Form of Fourier Series

Proof:

From (4.4) and Theorem 2.2 in Lecture 2, we easily see $c_0 = \frac{1}{2p} \int_{-p}^p f(t) e^{-i \frac{0\pi}{p} t} dt$

$$= \frac{1}{2p} \int_{-p}^p f(t) \cdot 1 dt$$

$$= a_0$$

For $n > 0$, using (4.1) we get

$$a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x$$

$$= a_n \frac{e^{i \frac{n\pi}{p} x} + e^{-i \frac{n\pi}{p} x}}{2} + b_n \frac{e^{i \frac{n\pi}{p} x} - e^{-i \frac{n\pi}{p} x}}{2i}$$

$$= \frac{1}{2} a_n e^{i \frac{n\pi}{p} x} + \frac{1}{2i} b_n e^{i \frac{n\pi}{p} x} + \frac{1}{2} a_n e^{-i \frac{n\pi}{p} x} - \frac{1}{2i} b_n e^{-i \frac{n\pi}{p} x}$$

$$= \frac{1}{2} (a_n + \frac{1}{i} b_n) e^{i \frac{n\pi}{p} x} + \frac{1}{2} (a_n - \frac{1}{i} b_n) e^{-i \frac{n\pi}{p} x}$$

$$= \frac{1}{2} (a_n + \frac{i}{i^2} b_n) e^{i \frac{n\pi}{p} x} + \frac{1}{2} (a_n - \frac{i}{i^2} b_n) e^{-i \frac{n\pi}{p} x}$$

$$= \frac{1}{2} (a_n - i b_n) e^{i \frac{n\pi}{p} x} + \frac{1}{2} (a_n + i b_n) e^{-i \frac{n\pi}{p} x}$$

Using the formulas for a_n and b_n from Theorem 2.2 in Lecture 2 and (4.4), we have

$$\frac{1}{2} (a_n - i b_n) = \frac{1}{2p} \int_{-p}^p f(t) \cos \frac{n\pi}{p} t dt - \frac{i}{2p} \int_{-p}^p f(t) \sin \frac{n\pi}{p} t dt$$

$$= \frac{1}{2p} \int_{-p}^p f(t) (\cos \frac{n\pi}{p} t - i \sin \frac{n\pi}{p} t) dt$$

$$= \frac{1}{2p} \int_{-p}^p f(t) e^{-i \frac{n\pi}{p} t} dt \quad \text{because of Euler's identity: } e^{-iu} = \cos u - i \sin u$$

$$= c_n$$

A similar argument shows that $c_{-n} = \frac{1}{2} (a_n + i b_n)$. Thus, for $n \geq 1$,

$$a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x = c_n e^{i \frac{n\pi}{p} x} + c_{-n} e^{-i \frac{n\pi}{p} x}$$

And so partial sum

$$S_N(x) = a_0 + \sum_{n=1}^N (a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x)$$

$$= c_0 + \sum_{n=1}^N c_n e^{i \frac{n\pi}{p} x} + \sum_{n=1}^N c_{-n} e^{-i \frac{n\pi}{p} x}$$

We change n to $-n$ in the second series on the right and combining, we have

$$S_N(x) = \sum_{n=-N}^N c_n e^{i \frac{n\pi}{p} x}$$

When $N \rightarrow \infty$, the theorem follows.

4.1 Complex Form of Fourier Series

The N th **partial sum** of (4.3) is by definition the **symmetric sum**

$$s_N(x) = \sum_{n=-N}^N c_n e^{i \frac{n\pi}{p} x} \quad (4.5)$$

We will see in a moment that is the same as the usual **partial sum** of the Fourier Series

$$f(x) = a_0 + \sum_{n=1}^N \left(a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x \right) \quad (4.6)$$

4.1 Complex Form of Fourier Series

If $h(x)$ is **complex-valued**, write $h(x) = u(x) + iv(x)$, where u and v are the real and imaginary parts of h . We define

$$\int h(x) dx = \int [u(x) + iv(x)] dx = \int u(x) dx + i \int v(x) dx \quad (4.7)$$

For example, if $h(x) = f(x)e^{iqx}$, where $f(x)$ is real-valued and q is a real number.

Then $h(x) = f(x)e^{iqx} = f(x)(\cos qx + i \sin qx)$. So

$$\int f(x)e^{iqx} dx = \int f(x) \cos qx dx + i \int f(x) \sin qx dx \quad (4.8)$$

where now **both integrals on the right are integrals or antiderivatives of real-valued functions.**

4.1 Complex Form of Fourier Series

It is straightforward to show that the integral is linear:

If h and g are complex-valued and a and β are complex numbers, then

$$\int (ah(x) + \beta g(x))dx = a \int h(x)dx + \beta \int g(x) dx \quad (4.9)$$

4.1 Complex Form of Fourier Series

We now highlight some interesting identities that relate the **complex Fourier coefficients** to the Fourier cosine and sine coefficients.

$$c_0 = a_0 \quad (4.10)$$

$$c_n = \frac{1}{2}(a_n - ib_n) \quad c_{-n} = \frac{1}{2}(a_n + ib_n) \quad (n > 0) \quad (4.11)$$

$$a_n = c_n + c_{-n} \quad b_n = i(c_n - c_{-n}) \quad (n > 0) \quad (4.12)$$

$$S_N(x) = s_N(x) \quad (4.13)$$

If f is real-valued, so that a_n and b_n are both real, (4.11) shows that c_{-n} is the complex conjugate of c_n . In symbols,

$$c_{-n} = \bar{c}_n \quad (4.14)$$

This identity fails in general if f is not real-valued.

4.1 Complex Form of Fourier Series

Example 4.1 A complex Fourier series

Find the complex form of the Fourier series of the 2π -periodic function

$f(x) = e^{ax}$ for $-\pi < x < \pi$, where $a \neq 0, \pm i, \pm 2i, \pm 3i, \dots$. Determine the values of the Fourier series at $x = \pm \pi$.

- Hyperbolic sine (双曲線正弦):

$$\sinh x = \frac{e^x - e^{-x}}{2} = \frac{e^{2x} - 1}{2e^x} = \frac{1 - e^{-2x}}{2e^{-x}}.$$

- Hyperbolic cosine (双曲線余弦):

$$\cosh x = \frac{e^x + e^{-x}}{2} = \frac{e^{2x} + 1}{2e^x} = \frac{1 + e^{-2x}}{2e^{-x}}.$$

- Hyperbolic tangent (双曲線正接):

$$\begin{aligned} \tanh x &= \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \\ &= \frac{e^{2x} - 1}{e^{2x} + 1} = \frac{1 - e^{-2x}}{1 + e^{-2x}}. \end{aligned}$$

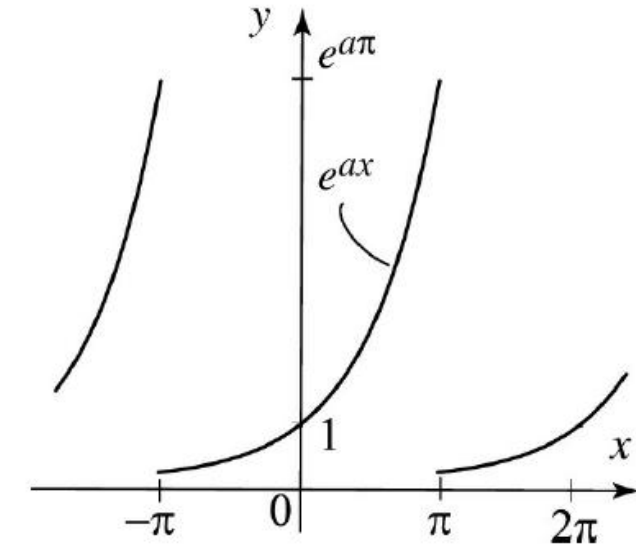
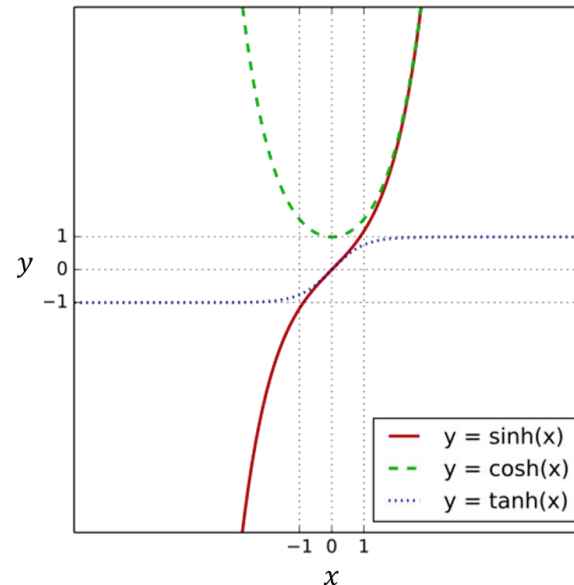


Figure 4.1 A 2π -periodic function, $f(x) = e^{ax}$, $-\pi < x < \pi$ where $a > 0$

4.1 Complex Form of Fourier Series

Solution:

From (4.4), we have

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ax} e^{-\frac{in\pi}{\pi}x} dx$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ax} e^{-inx} dx$$

$$= \frac{1}{2\pi} \left[\frac{e^{(a-in)x}}{a-in} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{2\pi} \left[\frac{e^{(a-in)\pi}}{a-in} - \frac{e^{(a-in)(-\pi)}}{a-in} \right]$$

$$= \frac{1}{2\pi} \cdot \frac{1}{a-in} \left[e^{a\pi} e^{-in\pi} - e^{-a\pi} e^{in\pi} \right]$$

$$= \frac{(-1)^n}{a-in} \frac{\sinh \pi a}{\pi}$$

where we have used $e^{\pm in\pi} = (-1)^n$
and $\sinh \pi a = \frac{e^{\pi a} - e^{-\pi a}}{2}$

Plugging this c_n into (4.3) and simplifying, we obtain the complex form of the Fourier series of f , i.e.

$$\frac{\sinh \pi a}{\pi} \sum_{n=-\infty}^{\infty} \frac{(-1)^n}{a-in} e^{inx} = \frac{\sinh \pi a}{\pi} \sum_{n=-\infty}^{\infty} \frac{(-1)^n}{a^2+n^2} (a+in) e^{inx}$$

Applying Theorem 4.1, we have

$$e^{ax} = \quad (-\pi < x < \pi)$$

Notice the discontinuous points $x = \pm\pi$, according to Theorem 4.1, they are given by the average of the function at these points.

From Figure 4.1, we see that this average is

$$\frac{e^{ax} + e^{-ax}}{2} = \cosh a\pi$$

4.1 Complex Form of Fourier Series

Orthogonality and Parseval's Identity

Similarly, the complex form of the Fourier coefficients can be obtained by appealing to **the orthogonality of the complex exponential system**

$$1, e^{i\frac{\pi}{p}x}, e^{-i\frac{\pi}{p}x}, e^{i\frac{2\pi}{p}x}, e^{-i\frac{2\pi}{p}x}, \dots, e^{i\frac{n\pi}{p}x}, e^{-i\frac{n\pi}{p}x}, \dots$$

The orthogonality of this system is expressed by

$$\frac{1}{2p} \int_{-p}^p e^{i\frac{m\pi}{p}x} e^{-i\frac{n\pi}{p}x} dx = \begin{cases} 0 & \text{if } m \neq n \\ 1 & \text{if } m = n \end{cases} \quad (4.15)$$

4.1 Complex Form of Fourier Series

Theorem 4.2 Complex Form of Parseval's Identity

Let f be a real-valued square integrable function on $[-p, p]$ with Fourier coefficients c_n given by (4.4). Then

$$\frac{1}{2p} \int_{-p}^p f^2(x) dx = \sum_{n=-\infty}^{\infty} |c_n|^2 \quad (4.16)$$

Proof:

From (4.10), we have $a_0^2 = c_0^2$.

With the help of (4.11) and (4.14), we obtain

$$\begin{aligned} |c_n|^2 &= c_n \bar{c}_n = \frac{1}{4} (a_n - ib_n)(a_n + ib_n) \\ &= \frac{1}{4} (a_n^2 + b_n^2) \end{aligned}$$

and similarly

$$\begin{aligned} |c_{-n}|^2 &= a_{-n} \bar{c}_{-n} = \frac{1}{4} (a_n + ib_n)(a_n - ib_n) \\ &= \frac{1}{4} (a_n^2 + b_n^2) \end{aligned}$$

Thus

$$\begin{aligned} \sum_{n=-\infty}^{\infty} |c_n|^2 &= |c_0|^2 + \sum_{n=1}^{\infty} |c_n|^2 + \sum_{n=1}^{\infty} |c_{-n}|^2 \\ &= a_0^2 + \frac{1}{4} \sum_{n=1}^{\infty} (a_n^2 + b_n^2) + \frac{1}{4} \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \\ &= a_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \\ &= \frac{1}{2p} \int_{-p}^p f^2(x) dx \end{aligned}$$

where the last equality follows from Parseval's identity (3.9) in Lecture 3.

*4.2 Uniform Convergence and Fourier Series

一様収束とフーリエ級数

* is optional. These contents will not be put into the final exam.

4.2 Uniform Convergence and Fourier Series

Review Lecture 1

EXAMPLE 1.3 Fourier series of the sawtooth function

The sawtooth function, shown in Figure 1.7, is determined by the formulas

$$f(x) = \begin{cases} \frac{1}{2}(\pi - x) & \text{if } 0 < x \leq 2\pi \\ f(x + 2\pi) & \text{otherwise} \end{cases}$$

- (a) Find its Fourier series.
- (b) Plot some **partial sums** and the Fourier series.

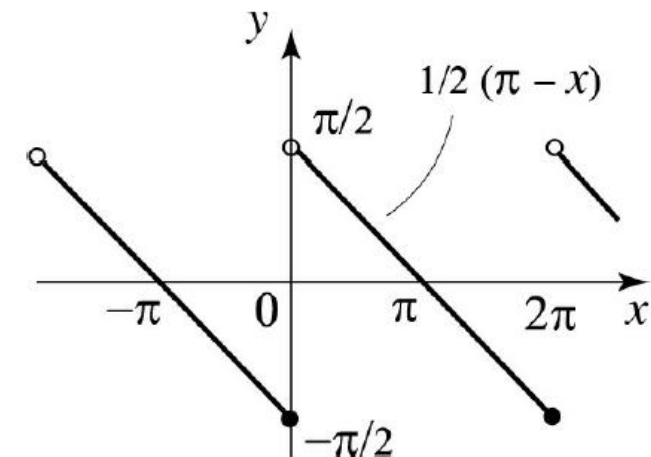


Figure 1.7 Sawtooth function.

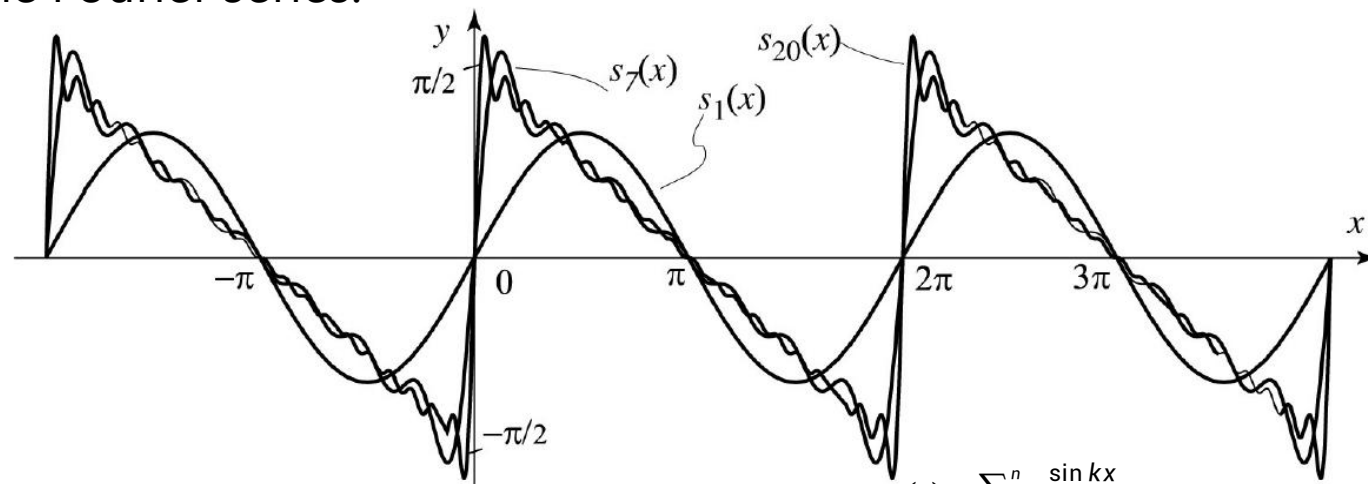


Figure 1.8 Here the n th partial sum of the Fourier series is $s_n(x) = \sum_{k=1}^n \frac{\sin kx}{k}$. distinguish the graphs, note that as n increases, the frequencies of the sine terms increase. This causes the graphs of the higher partial sums to be more wiggly.

Pointwise convergence

4.2 Uniform Convergence and Fourier Series

Review Lecture 2

EXAMPLE 2.2 Triangular wave

The 2π -periodic triangular wave is given on the interval $[-\pi, \pi]$ by

$$g(x) = \begin{cases} \pi + x & \text{if } -\pi \leq x \leq 0 \\ \pi - x & \text{if } 0 \leq x \leq \pi \end{cases}$$

(a) Find its Fourier series.

(b) Plot some **partial sums** and the Fourier series.

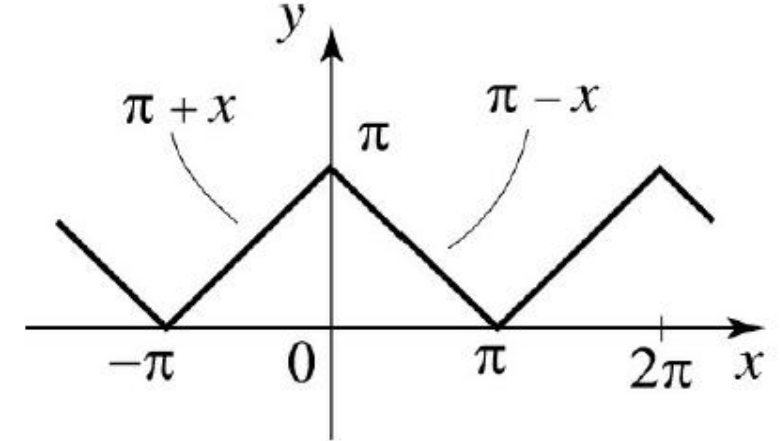


Figure 2.4 Triangular wave

Uniform convergence

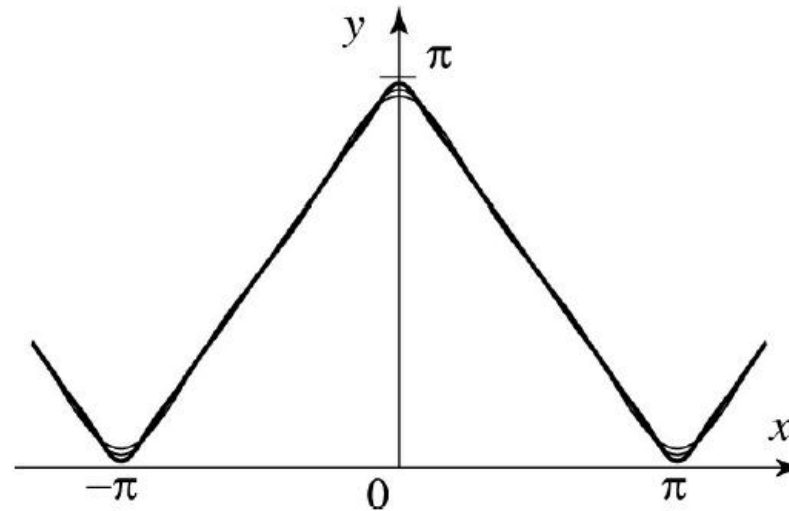


Figure 2.5 Partial sums of the Fourier series

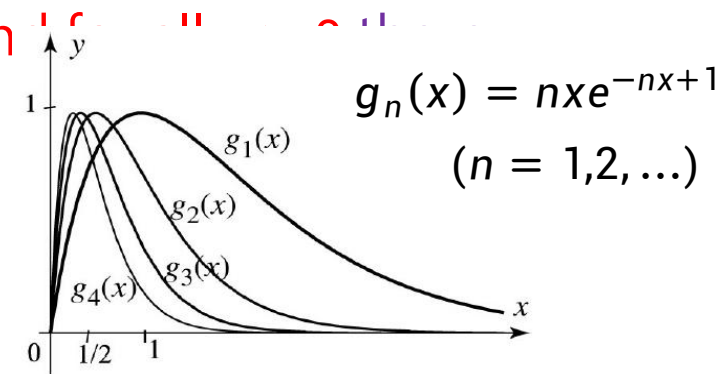
4.2 Uniform Convergence and Fourier Series

Pointwise Convergence and Uniform Convergence

- A sequence of functions $g_n(x)$, $n = 1, 2, \dots$, with common domain X is said to be

pointwise convergent to a limit function $g(x)$ if for all $x \in X$ and $\epsilon > 0$ there exists an $N \in \mathbb{N}$ such that if $n \geq N$ then:

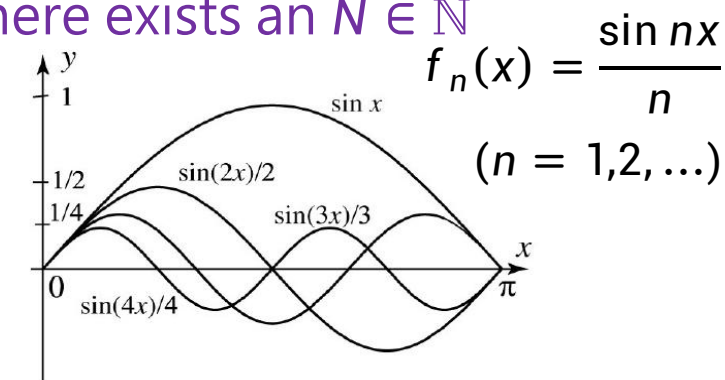
$$|g_n(x) - g(x)| < \epsilon$$



- A sequence of functions $f_n(x)$, $n = 1, 2, \dots$, with common domain X is said to be

uniformly convergent to a limit function $f(x)$ if for all $\epsilon > 0$ there exists an $N \in \mathbb{N}$ such that if $n \geq N$ we have that for all $x \in X$ that then:

$$|f_n(x) - f(x)| < \epsilon$$



4.2 Uniform Convergence and Fourier Series

The best way to understand the difference between these two definitions is to think of the definitions as a "game" process of sorts.

For **pointwise convergence**, think of it this way:

- You select an $x \in X$, and a friend gives you an $\epsilon > 0$.
- You then find an $N \in \mathbb{N}$ such that if $n \geq N$ we have that:

$$|g_n(x) - g(x)| < \epsilon$$

- If you succeed for each $\epsilon > 0$, then your friend asks you to choose another $x \in X$. You do, and once again, you show that for any $\epsilon > 0$ you can find such an N to satisfy the inequality above.
- For every $x \in X$ and for any $\epsilon > 0$ you can find such an $N \in \mathbb{N}$ then you will have shown that the sequence of functions $g_n(x)$, $n = 1, 2, \dots$, **converges pointwise** to the limit function $g(x)$.

In essence, the sequence of functions $g_n(x)$, $n = 1, 2, \dots$, converges pointwise to the limit function $g(x)$ if the sequence of numbers $g_n(x)$, $n = 1, 2, \dots$, converges to $g(x_0)$ for each $x_0 \in X$.

4.2 Uniform Convergence and Fourier Series

Now, for **uniform convergence**, think of it this way:

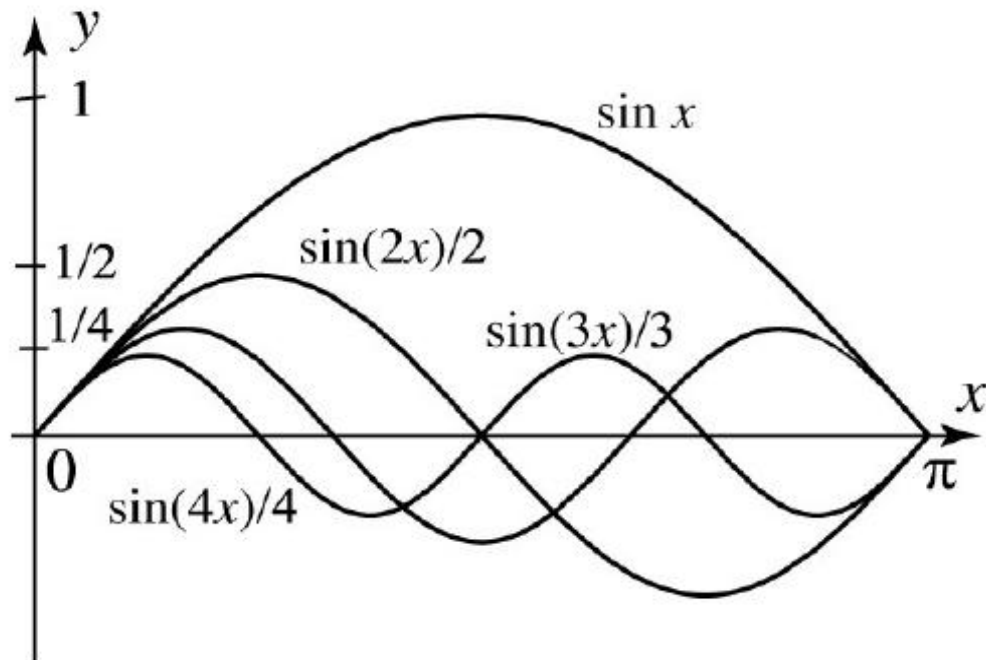
- Your friend gives you an $\epsilon > 0$.
- You must then find an $N \in \mathbb{N}$ such that if $n \geq N$ then for **EVERY** $x \in X$ we have that:

$$|f_n(x) - f(x)| < \epsilon$$

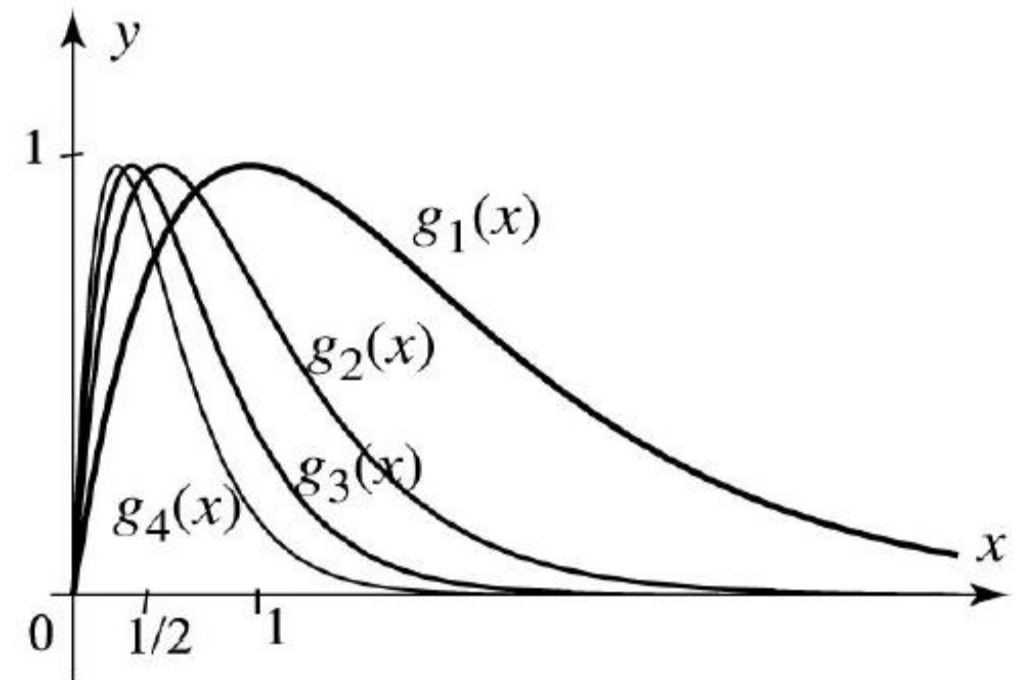
- If you succeed for each $\epsilon > 0$ then you will have shown that the sequence of functions $f_n(x)$, $n = 1, 2, \dots$, **converges uniformly** to the limit function $f(x)$.

In essence, the sequence of functions $f_n(x)$, $n = 1, 2, \dots$, converges uniformly to the limit function $f(x)$ if the sequence of numbers $f_n(x)$, $n = 1, 2, \dots$, converge to $f(x_0)$ for each $x_0 \in X$ at a somewhat similar/uniform rate.

4.2 Uniform Convergence and Fourier Series



(a) Uniform convergence



(b) Pointwise convergence

Figure 4.2 Uniform versus pointwise convergence

4.2 Uniform Convergence and Fourier Series

Theorem 4.3 Weierstrass M-Test

Let $u_k(x)$, $k = 0, 1, \dots$, be a sequence of real-valued or complex-valued functions on E .

Suppose that there is a sequence M_k , $k = 0, 1, \dots$, of nonnegative real numbers such that the following two conditions hold:

$$|u_k(x)| \leq M_k \quad \text{for all } x \text{ in } E. \quad (4.17)$$

and

$$\sum_{k=0}^{\infty} M_k < \infty \quad (4.18)$$

Then $\sum_{k=0}^{\infty} u_k(x)$ converges uniformly on E .

Remark: The Weierstrass M -test, when it applies, tells you that the series converges uniformly, but it does not give you the limit of the series.

4.2 Uniform Convergence and Fourier Series

Proof:

For $x \in E$, the series comparison test in Calculus I
and (4.18) show that the series

$$\sum_{k=0}^{\infty} u_k(x)$$

is (absolutely) convergent.

So we define a function u on E by

$$u(x) = \sum_{k=0}^{\infty} u_k(x).$$

Now, for all x in E , we have

$$\begin{aligned} \left| u(x) - \sum_{k=0}^n u_k(x) \right| &= |u_{n+1}(x) + u_{n+2}(x) + \dots| \\ &\leq |u_{n+1}(x)| + |u_{n+2}(x)| + \dots \quad (\text{triangle inequality}) \\ &\leq \sum_{k=n+1}^{\infty} M_k. \quad (\text{by (4.17)}) \end{aligned}$$

https://en.wikipedia.org/wiki/Triangle_inequality

the tail of a convergent series,

tends to zero as $n \rightarrow \infty$.

Consequently, given $\varepsilon > 0$, we can choose N

so that $|u(x) - \sum_{k=0}^n u_k(x)| < \varepsilon$ for all $n \geq N$,

which implies the uniform convergence of the series

4.2 Uniform Convergence and Fourier Series

Example 4.2 Weierstrass M -test

Show uniform convergence of the given series by using Weierstrass M -test

(a) The Fourier series in Example 2.2 of Lecture 2.

(b) The Series $\sum_{k=0}^{\infty} e^{-kx} \sin kx$, let $E = [1, \infty)$

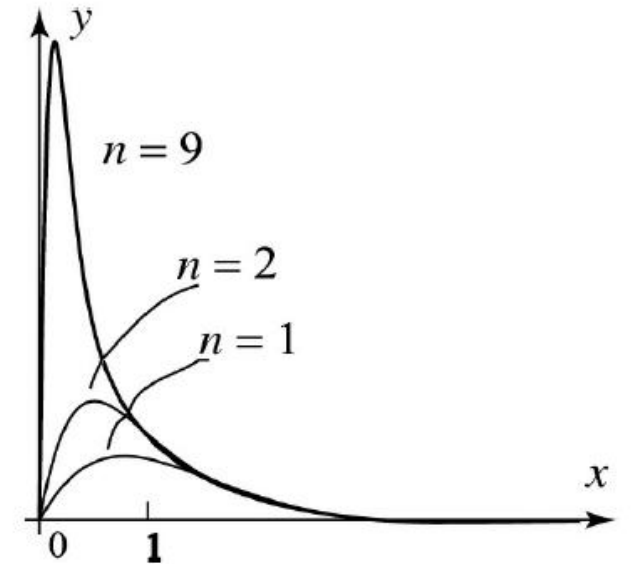


Figure 4.2 Uniform convergence of $\sum_{k=0}^{\infty} e^{-kx} \sin kx$ in $[1, \infty)$

4.2 Uniform Convergence and Fourier Series

Solution:

(a) The Fourier series of the function $g(x)$ in Example 2.2 of Lecture 2 is

$$\frac{\pi}{2} + \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\cos(2k+1)x}{(2k+1)^2}$$

Apply Weierstrass M-test with E equal the real line and $M_k = \frac{1}{(2k+1)^2}$. Since for all x in E

$$\left| \frac{\cos(2k+1)x}{(2k+1)^2} \right| \leq \frac{1}{(2k+1)^2}$$

$$\text{and } \sum M_k = \sum \frac{1}{(2k+1)^2} < \infty.$$

We then conclude from the Weierstrass M-test that the series converges uniformly on E .

The Fourier series representation theorem tell us that the limit of the series is the function $g(x)$.

Hence the series $\frac{\pi}{2} + \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{\cos(2k+1)x}{(2k+1)^2}$ converges uniformly to $g(x)$ on the entire real line.

(b) For all x in E , we have $|e^{-kx} \sin kx| \leq e^{-k} = M_k$.

Since

$$\sum_{k=0}^{\infty} e^{-k} = \frac{1}{1-e^{-1}} = \frac{e}{e-1} < \infty \quad (\text{geometric series})$$

the uniform convergence of the given series over the interval $E = [1, \infty)$ follows from the Weierstrass M-test (with $M_k = e^{-k}$).

4.2 Uniform Convergence and Fourier Series

Theorem 4.4 Continuity and Uniform Convergence

Suppose that f_n and u_k are continuous functions on E .

(a) If $f_n \xrightarrow{\infty} f$ uniformly on E , then f is continuous on E .

(b) If $\sum_{k=0}^{\infty} u_k$ converges uniformly to u on E , then u is continuous on E .

4.2 Uniform Convergence and Fourier Series

Corollary 4.1 Continuity of Fourier Series

Consider a $2p$ -periodic function f with Fourier series

$$a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi}{p} x + b_n \sin \frac{n\pi}{p} x \right)$$

- (i) If the Fourier series converges uniformly on E , then f must be continuous on E . Thus, continuity of f is necessary for uniform convergence.
- (ii) If the Fourier series converges uniformly on a closed interval of length $2p$, then it converges uniformly for all x . Moreover, f is continuous for all x .

4.2 Uniform Convergence and Fourier Series

Corollary 4.2 When Uniform Convergence Fails

Let f be a $2p$ -periodic piecewise smooth function. If f is not continuous at some point x , then its Fourier series does not converge uniformly on any interval that contains x .

4.2 Uniform Convergence and Fourier Series

Example 4.3 **Failure** of uniform convergence for the sawtooth function

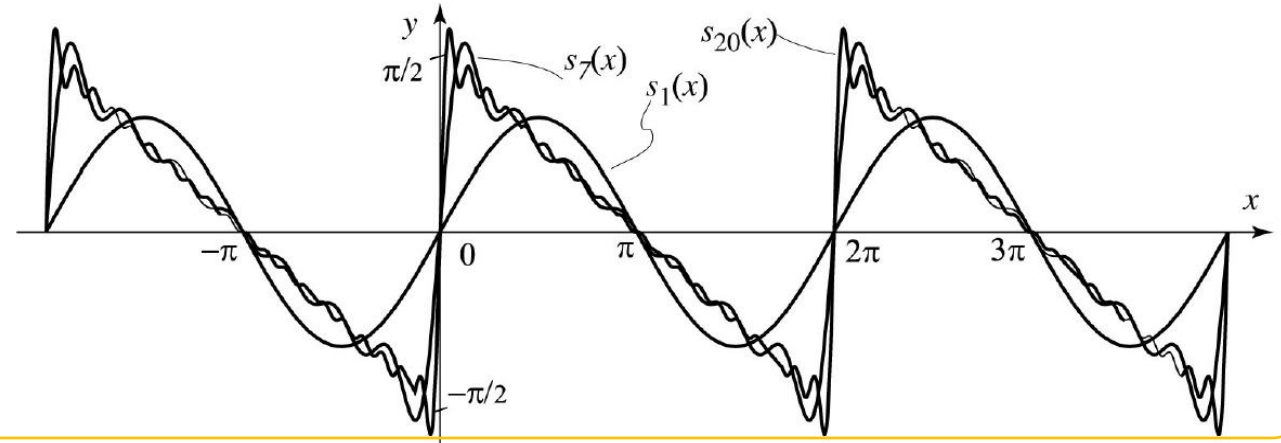
EXAMPLE 1.3 Fourier series of the sawtooth function

The sawtooth function, shown in Figure 1, is determined by the formulas

$$f(x) = \begin{cases} \frac{1}{2}(\pi - x) & \text{if } 0 < x \leq 2\pi \\ f(x + 2\pi) & \text{otherwise} \end{cases}$$

(a) Find its Fourier series.

(b) Plot some **partial sums** and the Fourier series.



Solution:

Since the sawtooth function (Example 1.3 in Lecture 1) is not continuous at the points $x = 2k\pi$ ($k = 0, \pm 1, \pm 2, \dots$), its Fourier series $\sum_{n=1}^{\infty} \frac{\sin nx}{n}$ does not converge uniformly on any interval containing any one of these points. In particular, the Fourier series does not converge uniformly on the interval $[0, 2\pi]$. (We note, however, that the Fourier series does converge uniformly on any closed interval that does not contain any one of the points $x = 2k\pi$. This is a more difficult result to prove.)

4.2 Uniform Convergence and Fourier Series

We now complete the picture by giving necessary and sufficient conditions for the uniform convergence of Fourier series.

Theorem 4.5 Uniform Convergence of Fourier Series

Suppose that f is piecewise smooth and $2p$ -periodic. Then the Fourier series of f converges uniformly to f on the entire real line if and only if f is continuous.

*4.3 Dirichlet Test and Convergence of Fourier Series

ディリクレの判定法とフーリエ級数の収束

* is optional. These contents will not be put into the final exam.

4.3 Dirichlet Test and Convergence of Fourier Series

Theorem 4.6 Dirichlet Test for Uniform Convergence

Let E be a subset of the real (or complex) numbers, $u_k(x)$, $k = 1, 2, \dots$ be a sequence of real- or complex-valued functions defined on E , and d_k , $k = 1, 2, \dots$ be a sequence of real numbers. Then, the series

$$\sum_{k=0}^{\infty} d_k u_k(x)$$

converges uniformly on E if the following two conditions are satisfied:

- (1) The coefficients d_k are positive and decreasing to zero.
- (2) There is a number M such that

$$\left| \sum_{k=0}^{\infty} u_k(x) \right| \leq M \quad \text{For all } x \text{ in } E \text{ and all } n$$

4.3 Dirichlet Test and Convergence of Fourier Series

Theorem 4.7 Trigonometric Series with Decreasing Coefficients

Suppose that a_k , $k = 1, 2, \dots$ and b_k , $k = 1, 2, \dots$ are sequences of positive numbers decreasing to zero.

(a) **(Uniform convergence)** Let $E = [a, b]$ where $0 < a < b < 2\pi$. Then the series $\sum_{k=1}^{\infty} a_k \cos kx$ and $\sum_{k=1}^{\infty} b_k \sin kx$ converge uniformly on E .

(b) **(Pointwise convergence)** The series $\sum_{k=1}^{\infty} b_k \sin kx$ converges point-wise for all x .

(c) **(Pointwise convergence)** The series $\sum_{k=1}^{\infty} a_k \cos kx$ converges for all x except possibly at the points $x = 2n\pi$, $n = 0, \pm 1, \pm 2, \dots$, where the series may converge or diverge.

4.3 Dirichlet Test and Convergence of Fourier Series

Example 4.4 Uniformly convergent trigonometric series

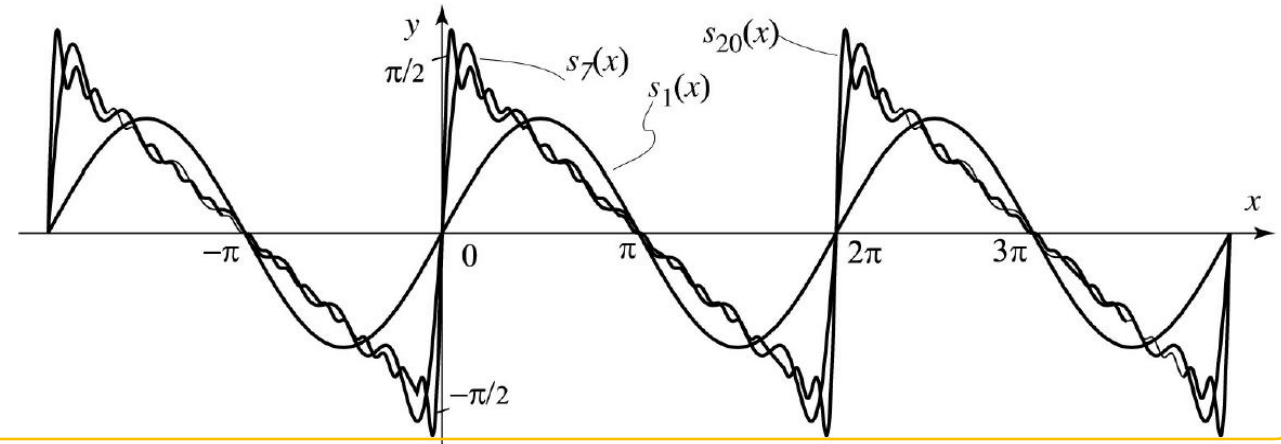
EXAMPLE 1.3 Fourier series of the sawtooth function

The sawtooth function, shown in Figure 1, is determined by the formulas

$$f(x) = \begin{cases} \frac{1}{2}(\pi - x) & \text{if } 0 < x \leq 2\pi \\ f(x + 2\pi) & \text{otherwise} \end{cases}$$

(a) Find its Fourier series.

(b) Plot some **partial sums** and the Fourier series.



Solution:

Straightforward applications of Theorem 4.7(a) show that the series $\sum_{k=1}^{\infty} \frac{\sin kx}{k}$, $\sum_{k=1}^{\infty} \frac{\sin kx}{\sqrt{k}}$ and $\sum_{k=1}^{\infty} \frac{\cos kx}{k \ln(k+1)}$ are uniformly convergent on the interval $[0.1, 6]$, or any other interval strictly contained in $[0, 2\pi]$. (Notice that none of these series can be handled by the Weierstrass M-test.)

Review for Lecture 4

- Complex Form of Fourier Series
- Orthogonality and Parseval's Identity
- *Pointwise Convergence and Uniform Convergence
- *Weierstrass ~~M~~Test
- *Dirichlet Test

Assignment

Please Check <https://web-ext.u-aizu.ac.jp/~xiangli/teaching/MA05/index.html>

Reading materials: Lecture 4 Slides Extension

References

[1] Nakhlé H. Asmar, *Partial Differential Equations with Fourier Series and Boundary Value Problems 2nd Edition*, 2004

[2] Wikipedia

4/20/2026